UNIVERSITY OF PERADENIYA
FACULTY OF SCIENCE

END-SEMESTER EXAMINATION, SEMESTER II- 2013/2014

MT 105 – REAL ANALYSIS I

Time Allowed: **Two hours** Answer all questions.

1. (a) Let A and B be non-empty subsets of $\mathbb{R}$. Denote $A+B=\{a+b \mid a\in A, b\in B\}$.
   Show that if A and B are bounded above, then $A+B$ is also a bounded above subset of $\mathbb{R}$.
   Establish the additive property, $\sup(A+B)=\sup A+\sup B$.

   (b) Let $a, b,$ and c be three rational numbers such that $a\sqrt{2}+b\sqrt{3}+c\sqrt{5}=0$. Find the values of $a, b,$ and c.

2. State the Monotone Convergence Theorem for sequences of real numbers.

   Let $1<a<3$ and let (x_n) be the sequence defined by $x_1=a$ and $x_{n+1}=\frac{1}{4}\left(x_n^2+3\right)$ for $n\in\mathbb{N}$. Show that (i) $1<x_n<3$ for $n\in\mathbb{N}$, and (ii) $x_{n+1}<x_n$ for $n\in\mathbb{N}$.

   Deduce that (x_n) is convergent and find $\lim\limits_{n\to\infty} x_n$.

   Hence, find $\lim\limits_{n\to\infty}\frac{x_n^3-x_n^2-2x_n+2}{x_n^2-1}$.

3. (a) Using the $\varepsilon-N$ definition, prove that $\lim\limits_{n\to\infty}\frac{2n^2+3n-1}{n^2+n-1}=2$.

   (b) Using the $\varepsilon-\delta$ definition, prove that $\lim\limits_{x\to 1}\frac{x^2+x+1}{x+1}=\frac{3}{2}$.

4. (a) Let $f(x)=\begin{cases} x & \text{if } x \text{ is rational,}\\ 0 & \text{if } x \text{ is irrational.}\end{cases}$

   Show that the function f is continuous at $a\in\mathbb{R}$ if and only if $a=0$.

   (b) Justifying the key steps, provide an example of a function $f:\mathbb{R}\to\mathbb{R}$ which is continuous only at the points 1 and 2.

[*Turn over*

5. (a) State *Rolle's Theorem.*

State and prove the *Mean-Value Theorem* in differential calculus.

Using the *Mean-Value Theorem*, show that $\frac{\pi}{4} - \frac{10}{181} < \tan^{-1}(0.9) < \frac{\pi}{4} - \frac{1}{20}$.

(b) Find each of the following limits:

(i) $\lim_{x\to 0} \frac{e^{3x} - e^{2x}}{e^{x} - e^{-x}} + \lim_{x\to\infty} x^2 e^{-x}$, (ii) $\lim_{x\to 0} \frac{\sin x - x\cos x}{x - \sin x}$, (iii) $\lim_{x\to 0} \frac{1}{x}\int_0^{2x} \cos(t^2)\,dt$